Element D

Ideas

Idea 1

Date: 2/17/25
Description: A model home that uses a flammable ceiling that holds back a mixture of sand and baking soda to help put out a fire. This system will use Arduino and temperature sensors to activate a buzzer in the event of a fire.

Idea 2

Date: 2/18/25
Description: Using an Arduino and a temperature sensor to activate a buzzer for notification and suppression using a portable fire-fighting foam spray that will be sprayed into the home using a servo motor activated after detection.

Idea 3

Date: 2/20/25
Description: To minimize false alarms, we will employ two sensors for fire detection: an MQ2 gas sensor and an infrared (IR) sensor. A buzzer will be utilized for notification purposes. Additionally, we will install tubes around the entire perimeter of the building to transport a mixture of water and baking soda from an external source, which will be circulated by a pump.

Idea 4

Date: 2/21/25
Description: We shall use the same plan from the last entry, but rather just use water instead of water and baking soda and water to minimize the cost and allow the water to flow easily through the tubes.

Decision Matrix

	weight	Option 1	Option 2	Option 3	Option 4

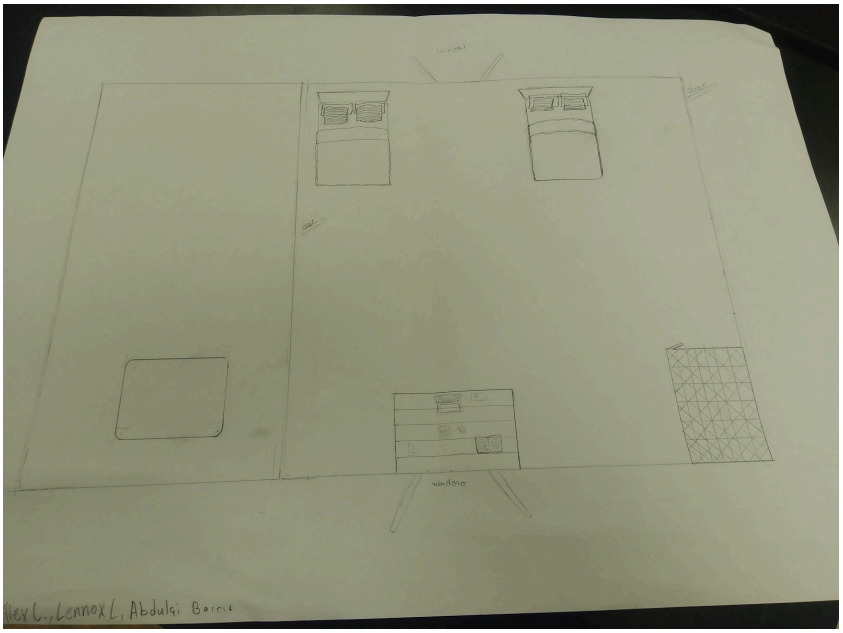
		Idea 1	Idea 2	Idea 3	Idea 4
Automation	1	1	1	1	1
Notification	2	2	2	2	2
Detection	3	1	1	3	3
Suppression	3	2	1	2	2.5
Cost	1	1	0.5	1	1
score	10	7	5.5	9	9.5

Our team chose design <u>idea 4</u>

Design Blueprint

Date 2/24/25

Sketch :



Manufacturing Methods:

Step 1: Rapid prototyping

Step 2: Machining

Step 3: Assembly

Step 4: Bolting

Design Blueprint

Date 2/28/25

Sketch: We added this feature outside the prototype to help hold the Arduino in place

Manufacturing Methods:

Step 1: Rapid prototyping

Step 2: Machining

Step 3: Assembly

Step 4: Bolting

<u>*NOTE* -</u> This design did not make it to the final product in order to prevent the electronics in Arduino from setting ablaze and causing more problems.